LocalMind demo and speaking guide

A practical handbook for a ten minute presentation

Use this guide to explain the project, run a controlled two minute demonstration, and answer viva questions. Read only the speech paragraphs aloud. Operation cues, preparation tasks, and optional answers are outside the timed script.

Local-First Agentic Knowledge Copilot with a Custom MCP Memory Server

SIT | Minors in AIML | Academic year 23-27
Guide Dr. Sonali Kothari

Presenters and running order

Presenter	PRN	Time
Trijal Burman	23070123145	00:00-02:15
Tiya Singh	23070123144	02:15-04:30
Gayatri Rajesh Apotikar	23070123055	04:30-06:30
Nimita Jestin	23070122259	06:30-10:00

Speaking roles organize the presentation; they do not assert individual implementation contributions. Gayatri operates the app during the demonstration.

Find what you need

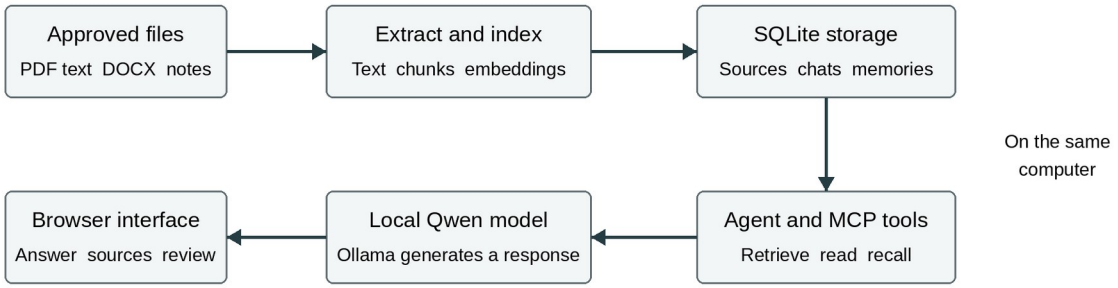
Pages	Use
2	Understand the system and explain RAG versus MCP
3-4	Prepare the app and follow the timed demonstration
5-8	Read the four verbatim speaking parts in order
9-11	Rehearse the 15 assigned viva answers
12	Check results, boundaries, and the final rehearsal

The ten minutes include the demo. Practise once with a timer; stop a slow model request after the demonstration wait limit and continue with the source and memory workflow.

Understand the project in plain language

LocalMind is a workspace for information already saved on a laptop. A user chooses folders on an HDD or SSD, indexes supported files, searches their contents, checks sources, and saves useful project context for later. The local model can help write an answer, summary, or draft plan, subject to its limits.

How a request moves through the system



The diagram shows the main data path. The browser and local API coordinate requests; SQLite supplies persistent records. The agent uses the custom MCP stdio server to reach approved knowledge and memory tools, and Ollama runs the downloaded model locally.

RAG and MCP solve different problems

RAG means retrieval augmented generation. Instead of relying only on a model's training, the application finds relevant passages and sends selected evidence with the question. Full-text search matches words; embeddings can help find related meaning. A retrieved fact can still be lost or misused before the final answer.

MCP means Model Context Protocol. It gives clients a common way to discover and call tools. Our server exposes local knowledge and memory functions. MCP does not rank passages by itself and is not a model or a vector database.

What memory means here

An explicit memory is a saved database record. A later request can retrieve it. Conversation history, extracted text, and plans are also stored locally. A model context window is temporary; saving a record does not train or fine-tune model weights, and every stored record is not sent on every turn.

The practical boundary

Initial software and model downloads require internet. Normal inference, indexing, and storage can then stay local without a required paid hosted API. Use a local MCP client throughout; a cloud-connected client could forward returned content. Local operation does not make generated answers automatically correct or encrypt the database.

Prepare the app before the clock starts

Finish setup and rehearsal beforehand. During the talk, show only the two supplied sample files. Do not connect a personal home folder or a whole drive for this demonstration.

Use a separate demo workspace

Start with a dedicated demo data directory, preserve the normal database, close private tabs, mute notifications, plug in the charger, and close unrelated GPU-heavy applications.

Check local model readiness

Open Settings > Local runtime and use Refresh status. Confirm the Chat model is Ready. Semantic search needs its embedding model; keyword search and stored text remain useful without it.

Connect the samples

Open Library > Connect folder. Enter the absolute path to examples/knowledge, label it Lighthouse demo, then choose Connect & index. Wait until indexing finishes and check any errors.

Prepare the project conversation

Create Lighthouse Presentation in Projects > New project. Use Open project chat. Start a new conversation if its scope is wrong; the project selector is disabled in an existing session.

Verify the source and rehearse once

Read lighthouse-project.md and locate Storage decision. Rehearse the exact question on page 4, but never describe a prepared response as a fresh live result. Keep the source reader available as a fallback.

Useful navigation

Task	Click path
Find a passage	Library > Search library > Search > Content matches > Read document
Read or export text	Read document > Full extracted text or Download text
Save a note manually	Memory > Add memory > Memory, Type, Scope > Save memory
Review project memory	Memory > project filter > Lighthouse Presentation

Current Linux sample folder: /run/media/trijal/HDD/SpecialisationBySol/examples/knowledge. On Windows, use the actual checkout folder followed by \examples\knowledge. The app address is normally http://127.0.0.1:8765; use the address shown by the launcher.

Run the two minute demonstration

Gayatri controls the keyboard from 04:30 to 06:30 of the full presentation.

Use these exact inputs

Question
What storage does the Lighthouse project use?

Explicit memory command
/remember Use only the supplied Lighthouse sample files in this presentation.

Expected source fact: SQLite. Verify it in lighthouse-project.md under Storage decision. This is a known sample fact, not a promise that the model will answer correctly.

After the restart on 17 October 2026, the two-document demo at http://127.0.0.1:8766 returned SQLite with source [1]. Search and recall appeared in its trace, and saved project memory remained available. This single rehearsal is not an accuracy benchmark.

Follow the sequence

Demo time	Operate
00:00-00:25	Library > lighthouse-project.md > Read document. Show Storage decision. Say G1 from page 7.
00:25-00:55	Close the reader. Projects > Lighthouse Presentation > Open project chat. Send the question once. Wait at most 10 seconds. If ready, open the source card and compare it. Say G2.
00:55-01:35	Send the /remember command. After Saved, open Memory and select Lighthouse Presentation. Point to the note. Say G3.
01:35-02:00	Reload Memory once. Show the saved note, stop clicking, and hand over to Nimita with G4.

If the model is slow or wrong

At ten seconds, move on. Show the indexed source directly. If a chat is still busy, save the note using Memory > Add memory and select Lighthouse Presentation under Scope. Use the accurate fallback language on page 7; do not repeat attempts until an answer looks good.

A page reload demonstrates that a saved record survives that reload. It is not a live application restart test. Storage tests provide separate persistence evidence.

Keep these outside the timed demo

No new folder scans, downloads, application restarts, long summaries, plan generation, or extra question attempts. Summary and plan actions are optional extensions only if the panel explicitly allows more time.

Trijal Burman

Verbatim script | 00:00-02:15

Read the regular text aloud. Italic operation cues are silent.

Operate T1 Show the title or app welcome screen. Face the panel.

Good morning, respected panel members. We are Trijal Burman, Tiya Singh, Gayatri Rajesh Apotikar, and Nimita Jestin from SIT. Under the guidance of Dr. Sonali Kothari, we are presenting LocalMind: a Local-First Agentic Knowledge Copilot with a Custom MCP Memory Server.

Operate T2 Keep the problem statement visible. Do not begin navigating.

Our problem is scattered knowledge and lost project context. A useful decision may be in a document, a lecture note, or an earlier conversation. Finding it again takes effort, and an ordinary chat has a limited context window. We wanted a workspace that helps a person find, inspect, and reuse their own approved information.

Operate T3 Point to Library, Memory, and Projects.

The prototype searches supported local files, shows source references, reads extracted document text, saves explicit memories, and drafts project plans. It uses existing local models rather than training a new large model. A saved memory belongs in a database; a relevant passage is supplied to the model when a request is made.

Operate T4 Show the scope or privacy note.

Local-first does not mean unrestricted drive access. The user chooses folders on the HDD or SSD. Initial software and model downloads need internet, but the intended workflow can then run offline without a paid hosted API. Hardware, electricity, and storage still have costs. Local processing also does not guarantee that every generated statement is correct.

Operate T5 Show the architecture on page 2. Hand over to Tiya.

Tiya will now explain how retrieval, persistent storage, and our custom MCP server work together. The speaking roles we have assigned today are presentation roles, not a claim about each person's individual implementation contribution.

Tiya Singh

Verbatim script | 02:15-04:30

Read the regular text aloud. Italic operation cues are silent.

Operate Y1 Trace approved files, extraction, and SQLite in the figure.

The workflow starts with an approved folder. Supported text is extracted and divided into smaller overlapping passages. SQLite stores the complete extracted text, passage information, conversations, and memories. EmbeddingGemma represents passages as numerical vectors, while SQLite full-text search finds matching words. The retrieval layer combines these signals to select useful evidence.

Operate Y2 Point from retrieved evidence toward the model.

RAG means retrieval-augmented generation. We retrieve evidence and provide selected passages with the question to a language model. Here, Qwen3 runs locally through Ollama. The response can contain numbered source references. Retrieval gives the model context, but we must still check whether the model received that context and used it correctly.

Operate Y3 Point to the MCP tool connection.

MCP has a different purpose: it is a standard interface for discovering and calling tools. Our custom local server provides search, document reading, memory recall, and project functions through a real stdio connection. RAG is the information-finding approach; MCP is the tool connection. It is not a model, database, or automatic accuracy improvement.

Operate Y4 Point to Memory and Projects.

An explicit memory persists in SQLite and can be retrieved in a later conversation. That is separate from the temporary model context and does not retrain its weights. Internal model-selected tools are read-only and bounded. The development setup used an RTX 5050, Ryzen 7 250, and sixteen gigabytes of RAM; that does not establish a universal performance guarantee.

Operate Y5 Return to Library. Give keyboard control to Gayatri.

Gayatri will demonstrate the prepared Lighthouse samples. We will check the actual source and save a memory, rather than treating a confident-looking response as proof of correctness.

Gayatri Rajesh Apotikar

Verbatim script | 04:30-06:30

Read the regular text aloud. Italic operation cues are silent.

Operate G1 D1 Open the sample source at Storage decision.

These are two supplied sample files, not personal documents. They were indexed before this timed demo. The Lighthouse document explicitly says to use SQLite for extracted text, conversations, and project memories. We can inspect that source directly.

Operate G2 D2 Ask the storage question once; use the 10 second limit.

I am asking a question whose answer is written in the sample. Once the response is ready, I compare it with its source card. If it disagrees or abstains, the document remains the evidence we should trust.

Operate G3 D3 Save the explicit memory; show its project scope.

Now I explicitly ask the application to save a demo preference. It appears in this project's Memory view. This save is an intentional database operation, not a claim that the language model has permanently learned the note.

Operate G4 D4 Reload Memory once; stop clicking and hand over.

The note is still listed after the page reloads. Persistence across application restarts is also covered by storage tests. Nimita will now explain what our measured results support and where the prototype still needs improvement.

Use the fallback that matches what happened

Model not ready or response exceeds ten seconds

The local model is not ready within our demonstration window. I will show the indexed source and saved memory instead. Generated answers and persistence are separate functions.

Wrong answer or false abstention

This response needs correction. The sample document explicitly says SQLite. We keep the source reader because a generated response can be wrong even when the file contains the answer.

No source card

This response did not return a usable source card. I will check the storage decision directly in Library.

If the app is unavailable, explain prepared artifacts as prepared. If private content appears, close the view or stop sharing before continuing. Never read private content aloud.

Nimita Jestin

Verbatim script | 06:30-10:00

Read the regular text aloud. Italic operation cues are silent.

Operate N1 Point to retrieval coverage in the results on page 12.

Our evaluation used authored fictional documents, not a representative personal archive. Hybrid retrieval included all required reference facts within its first five passages for ninety-six of ninety-six answerable retrieval questions. This is coverage on a small synthetic corpus, not one hundred percent answer accuracy. Lexical retrieval also reached the same coverage at five.

Operate N2 Point to the historical generation baseline: 0/24.

Historically, both generation conditions matched all requested values on zero of twenty-four answerable questions. The context fitter could discard retrieved evidence after serialization. An earlier evaluation-only diagnostic produced seven and three complete matches. That step is historical: the repair is now applied locally in the Linux and Windows checkouts, although it has not been committed or pushed.

Operate N3 Point to the production repair and direct replay.

The production repair fits a complete-character prefix using its actual serialized UTF-eight size. All sixty-four replay requests retained initial reference evidence, stayed within budget, and completed without request errors. Single-pass RAG matched all requested values on seven of twenty-four answerable questions; the agent did so on three. Average fact-value matching was thirty-seven point five and twenty-seven point zero eight percent. These are reused questions, not a held-out test or general accuracy. No extra model-selected tool calls occurred.

Operate N4 Point to historical summaries: 10/24; not rerun.

The earlier long-document experiment retained ten of twenty-four requested values across three final summaries, despite reading all extracted text. It was not rerun for this repair. A separate thousand-character passage-prefix limit also remains. Evidence retention and complete generated answers are still different properties.

Operate N5 Show the limits. Do not change model settings.

Other limits remain: no OCR for scanned image PDFs, readable rather than encrypted database storage, and access restricted to approved supported files. Linux inference was evaluated. The Windows copy passed tests on Linux, with native checks skipped in this repair run; native Windows GPU inference remains untested. Local operation does not remove every security risk.

Operate N6 Show future work, then return to the title or overview.

Next steps are better passage selection, stronger long-document coverage, broader held-out evaluation, and native Windows GPU testing. Local OCR is another possible extension. The context repair and regression tests are complete locally; these other improvements remain proposed. LocalMind combines a working local workflow with measurements of its remaining limits. Thank you to Dr. Sonali Kothari and the panel. We welcome your questions.

Viva questions one to five

The named presenter leads the answer; another teammate can add a verified detail.

1 What problem does the project solve

Answerer Trijal Burman

It helps a person find and reuse information from approved local files and saved project context. It combines search, source reading, explicit memory, and local model assistance. It does not automatically know an entire drive or guarantee every answer.

2 What is the difference between RAG and MCP

Answerer Tiya Singh

RAG retrieves evidence and supplies it to a generator. MCP standardizes how a client discovers and calls tools or reads resources. Our RAG workflow accesses local search and memory through MCP. MCP is not a language model, vector database, or ranking algorithm.

3 Explain the architecture

Answerer Tiya Singh

The browser calls a loopback FastAPI service. Approved files become text and overlapping chunks stored in SQLite. Full-text and embedding retrieval select passages. The agent calls the local MCP stdio server, then supplies bounded evidence to Qwen through loopback Ollama. The browser displays responses and sources.

4 Why a custom MCP server instead of only Python function calls

Answerer Tiya Singh

It provides a defined and reusable interface for knowledge and memory functions, including use by compatible local MCP clients. Our implementation uses a real protocol handshake. It adds overhead, so we do not claim MCP automatically improves retrieval accuracy.

5 What persists, and is that model memory

Answerer Gayatri Rajesh Apotikar

SQLite stores extracted text, chunks, conversations, and explicit memories. Plans are stored locally as Markdown. Later requests can retrieve that information. The model context is temporary, and saving records does not alter trained model weights or include every record in every prompt.

Viva questions six to ten

The named presenter leads the answer; another teammate can add a verified detail.

6 Does it have unrestricted access to all local files

Answerer Gayatri Rajesh Apotikar

No. The user approves folders. Unsupported formats, hidden items, known credential filenames, dependencies, links, and the app state folder are excluded. File-size and scan limits also apply. These checks reduce exposure but are not complete secret detection.

7 Did you train or fine-tune the language model

Answerer Nimita Jestin

No. We use existing downloaded Qwen3 and EmbeddingGemma models. The project concerns the application, retrieval, persistence, tool interface, and evaluation. Supplying retrieved context at request time is different from training or fine-tuning.

8 Which models and hardware did you use

Answerer Tiya Singh

The default chat tag is qwen3:4b-instruct-2507-q4_K_M, with EmbeddingGemma for 768-dimensional embeddings and an 8,192-token configured chat context. Development inference used Linux with an RTX 5050, Ryzen 7 250, and 16 GB RAM. We do not claim universal speed or native Windows GPU validation.

9 Is it completely offline and completely free

Answerer Trijal Burman

Normal use can run offline after the dependencies, runtime, and models are downloaded. There is no required paid hosted API. Hardware, power, storage, and initial downloads still cost resources. A cloud-connected external MCP client could also forward returned contents, so the client must be local to keep the full workflow local.

10 Does 96 out of 96 retrieval coverage mean 100% accuracy

Answerer Nimita Jestin

No. The frozen retrieval experiment put required evidence within five passages on 96 authored questions. Historical generation was 0/24 per method because the old fitter dropped evidence. The repaired production replay reached 7/24 single-pass RAG and 3/24 agent all-value matches on reused questions. These are distinct measurements, not universal answer accuracy.

Viva questions eleven to fifteen

The named presenter leads the answer; another teammate can add a verified detail.

11 What was the context bug, and is the fix deployed

Answerer Nimita Jestin

The old raw-prefix estimate missed actual JSON escaping and UTF-8 size, so a later check could discard the entire reference block. The production fitter now searches for a complete-character prefix that fits the actual serialized budget, including its shortening marker and metadata. The fix is applied locally to Linux and Windows copies, but not committed or pushed. Production replay retained initial references on 64/64 requests within budget. It reused the same questions, so it is not held-out validation.

12 Do citations prove factual correctness

Answerer Gayatri Rajesh Apotikar

No. A numbered citation can refer to a real document without supporting a particular claim. We inspect the cited passage. Automatic value matching and citation-index validation are useful proxies, not a complete factuality judgment.

13 Can it summarize every part of a long document

Answerer Nimita Jestin

No completeness guarantee. MCP pagination fetches supported extracted text and the agent processes long text in sections, but reduction can lose facts. The historical experiment retained 10 of 24 requested values across three final summaries; it was not rerun for this repair. The 1,000-character source-prefix limit for question answering also remains. The fix preserves a fitting reference message rather than redesigning retrieval or summary reduction.

14 What happens with scanned PDFs or without a model

Answerer Gayatri Rajesh Apotikar

This version does not OCR image-only PDF pages. Text-based PDFs, DOCX, text, Markdown, and supported code can be extracted. Without a chat model, indexing, full-text reading, keyword search, saved memory, and projects still work. Chat labels matching excerpts as extractive mode rather than pretending to generate a summary.

15 What are the main security, Windows, and future-work limits

Answerer Trijal Burman

The database is not encrypted, and local access protections do not defeat every local threat. The repair passed 62 Linux tests; the Windows copy tested on Linux passed 99 with six native checks skipped. Earlier native Windows CI is historical evidence, and Windows GPU inference remains untested. Priorities are passage coverage, held-out factuality evaluation, long summaries, and Windows GPU validation. The repaired context fitter and regression tests are already applied locally.

Results boundaries and final rehearsal

Keep retrieval, evidence transport, and final answer quality separate. These measurements use authored fictional English documents and automatic value matching; they are not a representative user study or a complete semantic factuality judgment.

Measurement	Observed result	What to say
Historical retrieval	96/96 at top five	Lexical and hybrid evidence coverage on the authored corpus.
Historical generation	0/24 per method	Original fitter dropped the reference block.
Repaired production	RAG 7/24; MCP 3/24	Direct replay; reused questions, not independent confirmation.
Evidence delivery	64/64 requests	Reference evidence retained within the configured budget.
Historical summaries	10/24 fact values	Three documents; not rerun after the repair.

Boundaries to state accurately

The app indexes approved supported folders rather than an unrestricted disk. Scanned image PDFs need OCR that this version does not include. Stored text is not encrypted. Model-selected tools are bounded and read-only. Linux inference was evaluated; native Windows GPU inference was not tested. Numbered sources must be checked against the claim.

Only if the panel asks for an extension

For a short summary, open the sample source and choose Summarize in chat. For a plan, open the project and choose Generate project plan with two milestones for the sample goals and a request to label assumptions. Both are reviewable generated drafts; they do not prove full summary coverage or create a finished software project.

Final rehearsal

- Check all four names and PRNs on page 1, then rehearse the handovers in order.
- Keep the total at ten minutes, including two minutes for Gayatri. Leave pauses around the measured results.
- The context repair is applied locally. Do not claim perfect answers, model training, agent superiority, a published repair, or tested Windows GPU inference.
- Next priorities: better passage selection, per-source evidence quotas, long-summary coverage, held-out factuality evaluation, native Windows GPU checks, and local OCR.